

PrintOps BMS Development Roadmap (Revised)

Goal

The objective is to reach a point where **the system can run a real print shop every day** without relying on external tools or manual bookkeeping. Every feature below is prioritized according to how much it affects daily operations.

Phase 0 — Stabilization (Highest Priority)

Before introducing new features, every existing feature must work reliably.

Objectives

- Fix Gold Meter (currently inactive)
- Fix Sale synchronization after payments
- Fix any remaining UI or navigation regressions
- Perform complete end-to-end workflow testing

Complete Workflow to Test

```
graph TD; Customer --> Proposal; Proposal --> ConvertToJob[Convert to Job]; ConvertToJob --> Production; Production --> Invoice; Invoice --> Payment; Payment --> Sale; Sale --> Dashboard; Dashboard --> Reports;
```

Customer
↓
Proposal
↓
Convert to Job
↓
Production
↓
Invoice
↓
Payment
↓
Sale
↓
Dashboard
↓
Reports

Every stage must correctly update the next one without requiring manual corrections.

Phase 1 — Complete the Core Job System

The **Job** should become the primary workspace for staff. Once a proposal is accepted, employees should rarely need to open the Invoice page.

1. Proposal → Job Transfer

When converting a proposal into a job, automatically transfer:

- Priority
- Assigned Staff
- Notes
- Customer information
- Services
- Quantities
- Pricing

No information should be lost during conversion.

2. Expand the Job Preview

The Job Preview should become the central management screen.

Instead of only displaying information, it should include quick actions such as:

- Edit Job
- Start Production
- Update Progress
- Generate Invoice
- Record Payment
- Print Job Sheet
- Cancel Job
- Duplicate Job

Users should not need to leave the preview just to make small edits.

3. Move Financial Information into Jobs

Currently financial information lives mostly inside Invoices.

Instead, every Job should clearly display:

Payment Summary

- Total Amount
- Amount Paid
- Remaining Balance
- Payment Status

Payment Status

- Unpaid
- Partial
- Paid
- Refunded (future)

Invoice Information

- Invoice Number
- Invoice Date
- Due Date

Quick Actions

- Add Payment
- View Payments
- Open Invoice
- Print Invoice

This allows staff to manage both production and payment from one screen.

4. Job Status Workflow

Current workflow should become standardized.

Draft

↓

Waiting

↓

In Production

↓

Finishing

↓

Ready for Collection

↓

Collected

↓

Completed

Special statuses:

- Cancelled
- On Hold
- Reprint

5. Production Progress

Continue improving the Progress Modal.

Include:

- Completed Quantity
- Remaining Quantity
- Reprints
- Assigned Staff
- Last Updated
- Estimated Completion

Phase 2 — Inventory Management

This becomes the second largest module.

Inventory should represent materials used in production.

Examples:

- Paper
- Vinyl
- Ink

- Toner
 - Lamination Sheets
 - Binding Materials
 - Packaging
-

Inventory Purchases

Inventory purchases should come from Expenses.

Example:

Expense

Category:

Inventory Purchase

↓

Vendor

↓

Material

↓

Quantity

↓

Cost

↓

Inventory increases

↓

Expense recorded

↓

Vendor balance updated

One action updates every related module automatically.

Inventory Features

Each inventory item should include:

- Material Name
 - Category
 - Unit
 - Quantity Available
 - Minimum Stock
 - Average Cost
 - Total Value
 - Supplier
 - Last Purchase Date
-

Stock Movements

Track every movement.

Stock In

Purchases

Stock Out

Production usage

Adjustments

Corrections

Waste

Damaged material

Nothing should change inventory without creating a movement record.

Phase 3 — Vendor Management

Vendor management should integrate tightly with Expenses.

Expense Form Improvements

Current expense entry should include a purchase type.

Expense Categories

- Utilities
- Salaries
- Maintenance
- Transport
- Marketing
- Petty Cash
- Inventory Purchase
- Vendor Payment

When selecting **Inventory Purchase** or **Vendor Payment**:

Display:

Vendor



Choose Existing Vendor

OR

Add New Vendor

After entering:

- Vendor
- Amount
- Description

the system should automatically:

- Create Expense
- Update Vendor Ledger
- Update Amount Owed
- Update Amount Paid
- Update Remaining Balance

No duplicate data entry.

Vendor Page

Each vendor should display:

Company Information

Purchase History

Outstanding Balance

Amount Paid

Remaining Balance

Payment Timeline

Recent Transactions

Materials Purchased

Expense History

This effectively becomes a supplier ledger.

Phase 4 — Machine & Service Redesign

Machines and Services should become independent.

Example

Machine

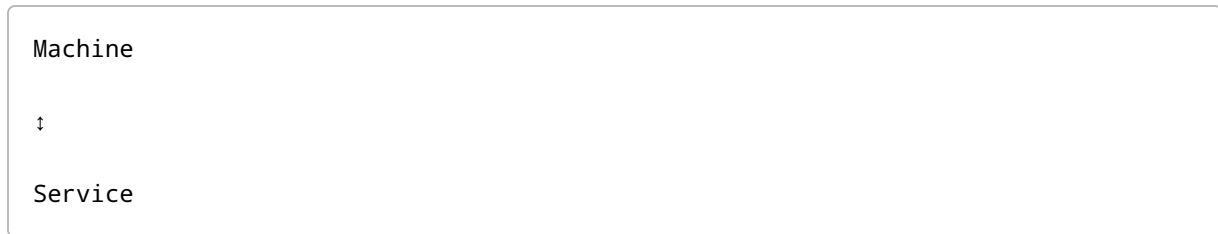
Konica Minolta

supports

- A4 Printing
- A3 Printing
- Booklets

while another machine may perform the same services.

Relationship:



Many-to-many.

This allows better production planning and redundancy.

Phase 5 — Financial Reporting

Improve financial accuracy.

Reports should distinguish between:

Sales

Expenses

Inventory Purchases

Vendor Payments

Profit

Outstanding Customer Balances

Outstanding Vendor Balances

Cash Flow

Monthly Profit

This provides a true picture of business performance.

Phase 6 — Analytics

Once operations are stable, improve decision making.

Examples:

- Daily Revenue
 - Monthly Revenue
 - Best Customer
 - Best Selling Service
 - Machine Utilization
 - Employee Productivity
 - Average Completion Time
 - Outstanding Jobs
 - Late Payments
 - Inventory Value
-

Future Features

These are useful but not essential for daily operation.

- User Roles & Permissions
 - Notifications
 - SMS & Email
 - Barcode Labels
 - QR Code Collection
 - Multi-Branch Support
 - Mobile Application
 - Cloud Synchronization
 - Customer Portal
-

Development Priority

Priority	Feature	Status
● P0	Fix Gold Meter	Pending
● P0	Fix Sale synchronization after payments	Pending
● P0	Full end-to-end workflow testing	Pending
● P1	Proposal → Job data transfer	Pending

Priority	Feature	Status
● P1	Expand Job Preview into the primary workspace	Pending
● P1	Move payment and invoice summaries into Jobs	Pending
● P1	Finalize production workflow and job statuses	In Progress
● P2	Inventory Management module	Not Started
● P2	Integrate Inventory with Expenses	Not Started
● P2	Vendor-integrated Expense system	Not Started
● P2	Vendor Ledger enhancements	In Progress
● P3	Machine ↔ Service redesign	Not Started
● P4	Financial reporting improvements	Not Started
● P5	Analytics dashboard	Not Started
● P6	Authentication, mobile app, notifications, multi-branch	Future

End Goal

At the completion of these phases, the system should allow a print shop to manage its entire business from a single platform:

- Customer Management
- Quotation & Proposals
- Job Management
- Production Tracking
- Invoicing
- Payment Collection
- Sales Recording
- Inventory Control
- Vendor & Supplier Management
- Expense Tracking
- Financial Reporting
- Business Analytics

No external spreadsheets or manual bookkeeping should be required for normal daily operations.